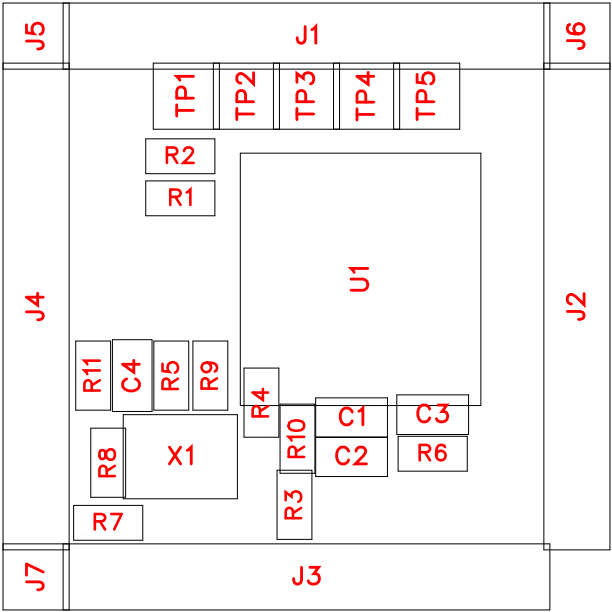
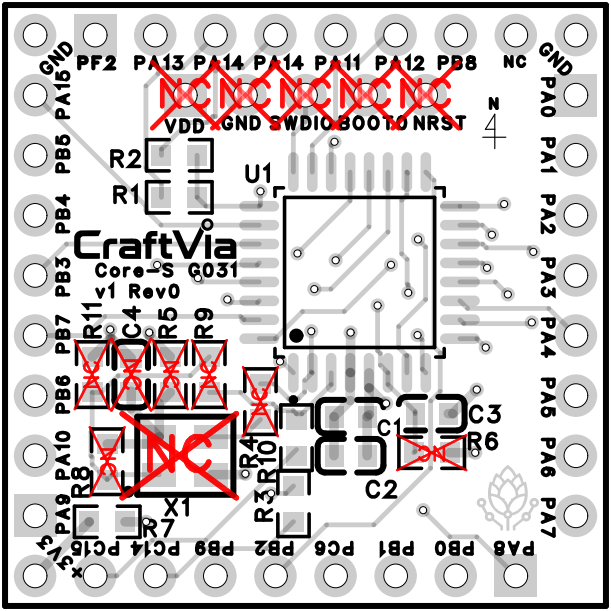

J5
J1
J6
TP1
TP2
TP3
TP4
TP5
R2
R1
U1
J4
J2
R11
C4
R5
R9
R4
C1
C3
R10
C2
R6
R8
X1
R3
R7
J7
J3

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier	LCSC Part Name
1	2	100nF	C1,C3	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	C14663	LCSC	100nF ±10% 50V
2	1	4.7uF	C2	C0603	4.7uF	CL10A475KO8NNNC	SAMSUNG(三星)	C19666	LCSC	4.7uF ±10% 16V
3	1	N 1-8	J1	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		C49256	LCSC	直排针 2.54mm 1*8P
4	1	E 1-8	J2	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		C49256	LCSC	直排针 2.54mm 1*8P
5	1	S 1-8	J3	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		C49256	LCSC	直排针 2.54mm 1*8P
6	1	W 1-8	J4	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		C49256	LCSC	直排针 2.54mm 1*8P
7	1	K-NW	J5	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	C81276	LCSC	1x1P 方针 直插
8	1	K-NE	J6	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	C81276	LCSC	1x1P 方针 直插
9	1	K-SW	J7	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	C81276	LCSC	1x1P 方针 直插
10	1	33R	R1	R0603	33Ω	0603WAF330JT5E	UNI-ROYAL(厚声)	C23140	LCSC	33Ω ±1% 100mW 厚膜电阻
11	1	1k	R2	R0603	1kΩ	0603WAF1001T5E	UNI-ROYAL(厚声)	C21190	LCSC	1kΩ ±1% 100mW 厚膜电阻
12	1	0R GPIO link	R3	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	C21189	LCSC	厚膜电阻 0Ω ±1%
13	1	0R / FB option	R7	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	C21189	LCSC	厚膜电阻 0Ω ±1%
14	1	0R	R10	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	C21189	LCSC	厚膜电阻 0Ω ±1%
15	1	STM32G031K8T6	U1	LQFP-32_L7.0-W7.0-P0.80-LS9.0-BL		STM32G031K8T6	ST(意法半导体)	C432203	LCSC	主流Arm Cortex-M0+ MCU，具有64 KB Flash存储器、8 KB RAM、64 MHz CPU、2x USART、定时器、ADC和通信接口，1.7-3.6V

Description	Mounting Type
Capacitance:100nF Tolerance:±10% Tolerance:±10% Voltage Rating:50V Temperature Coefficient:X7R	
Capacitance:4.7uF Tolerance:±10% Tolerance:±10% Voltage Rating:16V Temperature Coefficient:X5R	
Circluar Pins/Square Pins:- Current Rating (Max):- Voltage Rating (Max):- Length of Mating Pin:- Insulation Height:- Contact Material:	
Circluar Pins/Square Pins:- Current Rating (Max):- Voltage Rating (Max):- Length of Mating Pin:- Insulation Height:- Contact Material:	
Circluar Pins/Square Pins:- Current Rating (Max):- Voltage Rating (Max):- Length of Mating Pin:- Insulation Height:- Contact Material:	
Circluar Pins/Square Pins:- Current Rating (Max):- Voltage Rating (Max):- Length of Mating Pin:- Insulation Height:- Contact Material:	
Pin Structure:1x1P Pitch:- Row Spacing:- Circluar Pins/Square Pins:Pin Header Mounting Type:Through Hole Number of Rows:1 Number of Pins:1P Current Rating:3A Voltage Rating (Max):- Length of Mating Pin:6mm Insulation Height:2.5mm Length of End Connection	Through Hole
Pin Structure:1x1P Pitch:- Row Spacing:- Circluar Pins/Square Pins:Pin Header Mounting Type:Through Hole Number of Rows:1 Number of Pins:1P Current Rating:3A Voltage Rating (Max):- Length of Mating Pin:6mm Insulation Height:2.5mm Length of End Connection	Through Hole
Pin Structure:1x1P Pitch:- Row Spacing:- Circluar Pins/Square Pins:Pin Header Mounting Type:Through Hole Number of Rows:1 Number of Pins:1P Current Rating:3A Voltage Rating (Max):- Length of Mating Pin:6mm Insulation Height:2.5mm Length of End Connection	Through Hole
Type:Thick Film Resistor Resistance:33Ω Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:±100ppm/℃ Temperature Coefficient:±100ppm/℃ Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	

Type:Thick Film Resistor Resistance:1kΩ Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:±100ppm/℃ Temperature Coefficient:±100ppm/℃ Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
Type:Thick Film Resistor Resistance:0Ω Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:- Temperature Coefficient:- Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
Type:Thick Film Resistor Resistance:0Ω Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:- Temperature Coefficient:- Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
Type:Thick Film Resistor Resistance:0Ω Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:- Temperature Coefficient:- Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
CPU Core:ARM Cortex-M0+ CPU Maximum Speed:64MHz Core Size:32 Bit Program Storage Size:64KB Program Memory Type:FLASH RAM Size:8KB EEPROM:- Number of I/O:30 ADC (Bit):12bit Oscillator Type:Built-in+External Voltage - Supply:1.7V~3.6V Voltage - Supply:1.7V~	


GND
PF2
PA13
PA14
PA14
PA11
PA12
PB8
NC
GND
PA15
PA0
PB5
PA1
PB4
PA2
PB3
PA3
PB7
PA4
PB6
PA5
PA10
PA6
PA9
PA7
VDD
GND
SWDIO
BOOT0
NRST
R2
R1
U1
CraftVia
Core-S G031
v1 Rev0
R11
C4
R5
R9
R4
R10
C1
C3
R6
C2
R8
R3
X1
R7
VDD
PC15
PC14
PB9
PB2
PC6
PB1
PB0
PA8



CraftVia

CORE-S G031

MCU STM32G031K8T6

LOGIC 3.3 V

CPU 64 MHz / 64 KB FLASH / 8 KB RAM

FORMAT 1.0 IN / 2.54 MM PITCH

X1 OPTIONAL 8 MHz OSC (DNP)

CHECK ORIENTATION & PINOUT BEFORE INSERTION

DOCS craftvia.org

